SPECIFICATION SUBMISSION AND COMPLIANCE SHEET

Item : 11kv ACR (HS-8.400B)

Item No.	Technical Particulars	REB Specification	Guaranteed Specification
	Manufacturer's Name	To be mentioned	
	1. Design standard	ANSI C37.60	
	2. Catalog no. of proposed item.	To be mentioned	
	3. Country of origin.	To be mentioned	
	4. Installation Place	Outdoor, at 11 kv Feederof 33/11 Substations.	
H-8.400B	5. Environment of Installation Place	Temperatue- upto 65°c, Humidity up to 95% & Altitude upto 1,000 meters	
(400 Amps)	6. Circuit System	3 Phase, 11 KV, 50 Hz.	
	7 . Nominal System Voltage/Rated Max	11 KV/15 KV	
	8. Interrupting Medium	Vacuum, Vacuum Interrupters shall be in accordance with ANSI/IEEE C37.85	
	9. Control System	Electronic	
	10. Ground Trip Mechanism	Electronic	
	11. Insulating medium	Solid di-electric	
	12. BIL	110 KV	
	13. Low frequency voltage withstand	Dry-60 Sec : 50 KV	
		Wet- 10 Sec: 45KV	
	14. Fault type	The recioser shall be designd to sense line-to-ground, line-to-line or three phase fault current and shall interrupt all three phase of the distribution circuit simultaneously.	
	15. Ground/earth clearence (minimum)	350/520mm	
	16. Ground/earth creepage distence(minml	1180/671 mm	
	17. Rated Symetrical Interrupting current	12,000 Amps	
	18. Phase trip continuous current	400 Amps	
	19. Rated Min. phase tripping current	100-400 Amps	
	20. Rated Min. ground tripping current	10-400 Amps	
	21. Break Contact Structure	single break or double break	
	22. Number of fast curves	O to 4	
	23. Number of slow operation	Oto 4	
	24. Recloser head	Corrosion resistance Material	
	25. Recloser Tank	Corrosion resisting Steel	
	26. Connector	# 6 AWG to 350 MCM Cu or AL	
	27. Recloser Bushing	Silicon bushing with bushing clamps bolted to head with bird guard or Epoxy bushing and EPDM rubber bushing boots to provide a fully insulated bushing arrangement.	
	28. Lifting Provision	Shall be provided to lift with level	
	29. Manual Operation Provision	Shall be provided with manual operating lever	
	30. Lockout Provision	Non reclosing lever/feature to set recloser -to lockout after one operation.	
	31. Operation Counter	Shall indicate total operation	

Item No.	Technical Particulars	REB Specification	Guar�e!ed Specification
	32. Manual Closing tool	A manual closing tool shall be provided with each recloser.	
	33. Name Plate Information	Shall be provided as stated in the specification.	
	34. Current Transformer	Recloser shall be equipped with multi-ratio CT for metering & protection. CT shall cover the entire r;,nn<>	
	35. Metering Provision	Recloser shall be provided with 3 phase Multifunction meter for reading **A, V, *'t'0N,* KVAR, KWH R** & P.F.	
	36. Protection Settings	Current, time and related operational settings shall be programmable through the self-contained built-in keyboard & display of the Control cubical of the ACR itself. Protection setting through the full range of the device should be programmable without changing the CTs.	
	37. Co-ordination capability (TCC Curve)	Class III 11 KV reclosure shall have dual time current capability for field adjustment and selection of alternate curves of Standard inverse, Very inverse or extremely inverse.	
H-8.400B (400 Amps)		The recloser shall be capable of to co-ordinate with 6.35 KV line reclosure, 33 Kv power fuses, 33 KV 3 pase reclosure and must be adjustable to trip to lock out later than the single phase line reclosures, which may be set for two fast curve and two delayed curve operations to lockout. Between the fast and slow operations there may be sufficient time delay to allow for operation of fuses.	
	38. Factory Setting for Phase and Ground trip operating sequences	Shall be as given in Table 1 of specification	
	39. Provision for field adjustment	No. of fast operation: - one, two, three or four.	
		Total operation to lock-out :-two, three or four	
		Non-reclosing features-Operable by external lever or provision in electronic control to provide one operation to lock-out without reclosIng.	
	40. Reclosing Time Interval	Adjustable 0.6 to 2.5 Sec with 0.1 sec. resolution	
	41. Reset Time	Variable from O sec. to 10 seconds.	
	42. Prov. for different trip current setting	Without changing CTs	
	43. Substation mounting frame	Shall be provided	
	44. Auxilllary power supply	The reclosure shall be provided with auxiliary transformer, Battery & Charger of required rating to provide power supply to the control panel. The PT output must be 240 V.	
	45. List of spare parts	List of manufacturer's recommended spare parts for 2 v<>,.rc; with orice shall hP submittP<I.	
	46. Design & Product Test	Different Tests as mentioned in the specification shall be carried out as per ANSI C37.60 & relevant standard from Internationally recognized independent testing laboratory and shall be submitted with the bid proposal.	
	47. Warranty	The bider shall warrant that the ACR furnished have conformed to REB specification. Within 3 Years from the date of delivery if the ACR is found to have defects, the bidder shall repair or replace the defective parts at free of cost.	
	48. Packing	As per shipping box design no. 2 of exhibit H-2 of tender document	

SPECIFICATION SUBMISSION AND COMPLIANCE SHEET

Item : 33 kv ACR (HS-7.800B)

Item No.	Technical Particulars	REB Specification	Guaranteed Sceclfication
HS-7.8008	Manufacturer's Name	To be mentioned	
	1. Desian standard	ANSI C37.60	
	2. Cataloq no. of orooosed item.	To be mentioned	
	3. Countrv of oriain.	To be mentioned	
	4. Installation Place	At 33 KV Incomimq lines of REB S/S	
	5. Environment of Installation Place	Temperatue- upto SO^{0}C, Humidity up to 95% & Altitude upto 1,000 meters	
	6. Circuit System	3 Phase 33 Kv 50 Hz	
	7. Nominal System Voltage/Rated Max voltaae	33 KV/38 KV	
	8. Interrupting Medium	Vacuum, Vacuum Interrupters shall be in ;:irmrrl;:ince with ANSI/IEEE C"17.85	
	9. Control Svstem	Electronic	
	10. Ground Trio Mechanism	Electronic	
	11. Insulatina medium	Solid-Dielectric	
	12. BIL	150 KV	
	13. Low frequency voltage withstand	Drv-60 Sec : 70 KV	
		Wet- 10 Sec: 60KV	
	14. Fault type	The recloser shall be designd to sense line-to-ground, line-to-line or three phase fault current and shall interrupt all three phase of the distribution circuit simultaneously.	
	15. Ground/earth clearence (minimum)	394/676mm	
	16. Ground/earth creepage distance (minm)	1763/1215 mm	
	17. Rated Symetrical Interrupting 1current	16000 Amps	
	18. Rated Min. ohase triooina current	100-800 Amos	
	19. Rated Min. ground tripping current	10-800 Amps	
	20. Break contact structure	sinale break or double break	
	21. Number of fast curve	Oto 4	
	22. Number of Slow ooeration	Oto 4	
	23. Recloser head	Corrosion resistance material	
	24. Recloser Tank	Corrosion resisting steel	
	25. Connector	# 6 AWG to 350 MCM Cu or AL	
	26. Recloser Bushing	Wet process porecelain with bushing clamps bolted to head with bird (guard) protection or Cyclosphatic bushing and EPDM rubber bushing boots to orovide a fully bushing arrangement.	
	27. Liftinq Provision	Shall be provided to lift with level	
	28. Manual Ooeration Provision	Shall be orovided with manual operating lever	
	29. Lockout Provision	Non reclosing lever/feature to set recloser to lockout after one operation.	
	30. Ooerations Counter	Shall indicate total ooeration	
	31. Manual Closing tool	A manual closing tool shall be provided with each recloser.	
	32. Name Plate information	Shall be provided as stated in the specification.	
	33. Cable for control / meter box	10 meter long contol cable shall be provided	
	34. Current Transformer	Recloser shall be equipped with multi-ratio CT for metering & protection. CT shall cover the entire protection range.	
	35. Metering Provision	Recloser shall be provided with 3 phase Multifuction meter for reading A, V, KW, KVAR, KWH R& P.F.	

Item No.	Technical Particulars	REB Specification	**Guaranteed Soecification**
HS-7.8008	36. Protection Settings	Current, time and related operational settings shall be programmable through the self-contained built-in keyboard & display of the Control cubical of the ACR itself. Protection setting through the full range of the device should be programmable without changing the CTs.	
	37. Co-ordination Capability (TCC Curve)	Class III 33 KV reclosure shall have dual time current capability for field adjustment and selection of alternate curves of Standard inverse, Very inverse or extremely inverse.	
		The recloser shall be capable of to co-ordinate with 100E/125E/175E standard speed power fuse at 33 KB Bus and must be adjustable to trip to lock out later than the maximum clearing time of the indicated power fuse and faster than the damage curve of 33 Kv side conductor as well as the 33/11 Power transformer of the 5/5.	
	38. Factory Setting for Phase and Ground trio OnPratina seauences	Shall be as given in Table 1 of specification	
	39. Provision for field adjustment	No. of fast operation: - one, two, three or four.	
		Total operation to lock-out :-two three or four	
		Non-reclosing features-Operable by external lever or provision in electronic control to provide one operation to lock-out without reclosing.	
	40. Reclosing Time Interval	Adjustable 0.6 to 2.5 Sec with 0.1 sec. resolution	
	41. Reset Time	Variable from 20 sec. to 120 seconds.	
	42. Prov. for different trip current i=ttina	Without changing CTs	
	43. Pole mounting hanger:	Pole mounting hanger shall be provided.	
	44. Auxilliary power supply	The reclosure shall be provided with auxiliary transformer, Battery & Charger of required rating to provide power supply to the control panel. The PT output must be 240 V	
	45. List of spare parts	List of manufacturer's recommended spare parts for 2 years with price shall be submitted.	
	46. Design & Product Test	Different Tests as mentioned in the specification shall be carried out as per ANSI C37.60 & relevant standard from Internationally recognized independent testing laboratory and shall be submitted with the bid proposal.	
	47. Warranty	The bider shall warrant that the ACR furnished have conformed to REB specification. Within 3 Years from the date of delivery if the ACR is found to have defects, the bidder shall repair or replace the defective parts at free of cost.	
	48. Packing	As per shipping box design no. 2 of exhibit H-2 of tender document	

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